THE MAP ON DIFFERENTIALS FOR A SURJECTION, WITH ITS KERNEL IDENTIFIED

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The map of presentations induced by a square of ring maps. Used at the start of the counted proof (lines 34204-34207). Gives the surjectivity of the map on differentials from the surjectivity of the top map of free modules.
- Description of differentials of the target by generators and relations. Used in the kernel computation (lines 34208-34233). Lets an element mapping to zero be written in a normal form that exposes its coefficients.

Source reference. Original equation 00RQ: sources/stacks-algebra/pinned-original-source.tex, source lines 34114–34114. Navigation pointer for \ref{equation-functorial-omega} used inside the counted span (line 34196); the argument that consumes it is retained in the span.

Source reference. Original equation 0H2F: sources/stacks-algebra/pinned-original-source.tex, source lines 34144–34144. Navigation pointer for \ref{equation-map-presentations} used inside the counted span (line 34205); the argument that consumes it is retained in the span.

Date: stacks-project:algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *In diagram (00RQ), suppose that $S \to S'$ is surjective with kernel $I \subset S$. Then $\Omega_{S/R} \to \Omega_{S'/R'}$ is surjective with kernel generated as an S-module by the elements da, where $a \in S$ is such that $\varphi(a) \in \beta(R')$. (This includes in particular the elements $d(i)$, $i \in I$.)*

First proof. Consider the map of presentations (0H2F). Clearly the right vertical map of free modules is surjective. Thus the map is surjective. Suppose that some element η of $\Omega_{S/R}$ maps to zero in $\Omega_{S'/R'}$. Write η as the image of $\sum s_i[a_i]$ for some $s_i, a_i \in S$. Then we see that $\sum \varphi(s_i)[\varphi(a_i)]$ is the image of an element

$$\theta = \sum s'_j[a'_j, b'_j] + \sum s'_k[f'_k, g'_k] + \sum s'_l[r'_l]$$

in the upper left corner of the diagram. Since φ is surjective, the terms $s'_j[a'_j, b'_j]$ and $s'_k[f'_k, g'_k]$ are in the image of elements in the lower right corner. Thus, modifying η and θ by subtracting the images of these elements, we may assume $\theta = \sum s'_l[r'_l]$. In other words, we see $\sum \varphi(s_i)[\varphi(a_i)]$ is of the form $\sum s'_l[\beta(r'_l)]$. Next, we may assume that we have some $a' \in S'$ such that $a' = \varphi(a_i)$ for all i and $a' = \beta(r'_l)$ for all l. This is clear from the direct sum decomposition of the upper right corner of the diagram. Choose $a \in S$ with $\varphi(a) = a'$. Then we can write $a_i = a + x_i$ for some $x_i \in I$. Thus we may assume that all a_i are equal to a by using the relations that are allowed. But then we may assume our element is of the form $s[a]$. We still know that $\varphi(s)[a'] = \sum \varphi(s'_l)[\beta(r'_l)]$. Hence either $\varphi(s) = 0$ and we're done, or $a' = \varphi(a)$ is in the image of β and we're done as well. $\square$